SUCCESSIONI

↓

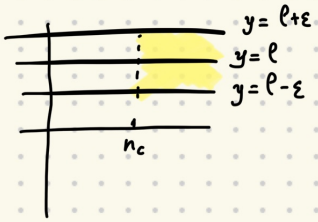
funzione $a: \mathbb{N} \to \mathbb{R}$ $a(n)$ oppure a_n

- COSTANTI $a_n = c \ \forall n$, $a_n = 2n \ \forall n$, $a_n = 2n+1 \ \forall n$

- CONVERGENTE

UNA SUCCESSIONE a_n È CONVERGENTE A $\ell \in \mathbb{R}$

E $\forall \varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N} : |a_n - \ell| < \varepsilon \ \forall n > n_\varepsilon$



$a_n = \frac{1}{n+1} \to 0$

UNA SUCCESSIONE CONVERGENTE È LIMITATA

DIM $\varepsilon = 1$ ℓ = LIMITE

$\{a_n : n \geq n_\varepsilon\} \subseteq [\ell - 1, \ell + 1]$

$\{a_0, a_1, a_{\varepsilon-1}\} \subseteq [m, M]$

PROP SE a_n CONVERGE IL LIMITE È UNICO

DIM SUPPONIAMO CHE $a_n \to \ell$, $a_n \to \ell'$

$|\ell - \ell'| = |(\ell - a_n) + (a_n - \ell')| = |a_n - \ell| + |a_n - \ell'| \leq \underbrace{\varepsilon + \varepsilon}_{2\varepsilon}$

$\forall \varepsilon > 0 \ \exists n_\varepsilon : |a_n - \ell| < \varepsilon \ \forall n \geq n_\varepsilon$

$\exists n'_\varepsilon : |a_n - \ell'| < \varepsilon \ \forall n \geq n'_\varepsilon$

$N_\varepsilon = \max\{n_\varepsilon, n'_\varepsilon\}$

$|\ell - \ell'| \leq 2\varepsilon \ \forall \varepsilon > 0 \Rightarrow \ell = \ell'$

PROPRIETÀ DEI LIMITI

$a_n \to L$ $b_n \to M$

I. $a_n + b_n \to L + M$
II. $a_n - b_n \to L - M$
III. $a_n \cdot b_n \to L \cdot M$ DIM
IV. $M \neq 0$ $\frac{a_n}{b_n} \to \frac{L}{M}$

$|a_n b_n - LM| = |M(L - a_n) + a_n(M - b_n)|$ → Abbiamo aggiunto $a_n M$

$= |ML - Ma_n + Ma_n - a_n b_n|$

$\forall \varepsilon > 0 \ \exists n_\varepsilon : |a_n - L| < \varepsilon \ \forall n > n_\varepsilon$,

$|b_n - M| < \varepsilon \ \forall n > n_\varepsilon$

$\leq |M(L - a_n)| + |a_n(M - b_n)|$

$\leq |M| |L - a_n| + |a_n| |M - b_n|$

$\leq |M| \varepsilon + |a_n| \varepsilon \leq \varepsilon (|M| + K) \to 0$

$a_n \to L \Rightarrow |a_n| \leq K \ \forall n$

$\to |a_n b_n - LM| \leq (|M| + K)\varepsilon \to 0$

ESEMPIO

$$\lim_{n\to\infty} \frac{2n^2-1}{3n^2+n+1} = \lim_{n\to\infty} \frac{n^2\left(2-\frac{1}{n^2}\right)}{n^2\left(3+\frac{1}{n}+\frac{1}{n^2}\right)} = \frac{2}{3}$$

($2 - \frac{1}{n^2} \to 2$; $3 + \frac{1}{n} + \frac{1}{n^2} \to 3$)

TEOREMA DEI CARABINIERI

$a_n \le b_n \le c_n \qquad \forall n \ge N$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = \ell$

$b_n \to \ell$

DIM

$\forall \varepsilon > 0 \ \exists n_\varepsilon : |a_n - \ell| < \varepsilon \qquad \forall n \ge n_\varepsilon$

$\ell - \varepsilon < a_n < \ell + \varepsilon$

$\ell - \varepsilon < c_n < \ell + \varepsilon$

$\underbrace{\ell - \varepsilon < a_n \le b_n \le c_n < \ell + \varepsilon}_{\Downarrow} \qquad \forall n \ge n_\varepsilon$

$|b_n - \ell| < \varepsilon$